Sahil Khunt

skhun@illinois.edu | sahilkhunt08.github.io/portfolio | linkedin.com/in/sahilkhunt | (224) 239-8445

Education

University of Illinois Urbana-Champaign GPA: 3.79/4.00
Bachelor of Science in Computer Science, Minor in Statistics *Expected May 2028*

Relevant Coursework: Data Structures, Discrete Structures, Multivariable Calculus, Linear Algebra, Algs & Models of Comp, Database Systems, Computer Architecture, System Programming

Experience

Wells Fargo - *Software Engineer Intern* June 2026 – Aug 2026
- Built a Python/Streamlit QA automation platform used by 130+ developers, automating enterprise REST API validation and regression testing across development and test environments.
- Designed a MongoDB-backed test data management system to persist and reuse generated test data, reducing redundant ingestion and improving regression efficiency.
- Modernized legacy testing workflows by replacing deprecated APIs and removing obsolete configurations.

Disruption Lab - *Technical Lead* Sept 2025 – May 2026
- Led 5 SWEs in designing and building a multi-stage Python pipeline for an enterprise tax compliance client, using lxml and XPath to parse XSD schemas and detect year-over-year tax form changes.
- Automated comparisons across nearly 4K pairs of annual state tax-form XSD files with a 98% success rate, detecting 11.5K+ element-level additions, removals, and modifications.
- Integrated an LLM analysis layer to classify XSD changes, summarize impact, and recommend validation tests.

Social Computing Systems Lab - *Software Engineering Researcher* Sept 2025 – Jan 2026
- Developed an AI-powered research networking platform for personalized researcher discovery and matching.
- Built multi-stage Selenium pipelines to scrape and structure 10K+ publications across 4 research conferences.
- Implemented Dify-based semantic retrieval with PostgreSQL/Drizzle for tailored researcher recommendations.

FiPet - *Software Engineer Intern* May 2025 – Aug 2025
- Built and shipped features for FiPet's financial literacy app using React Native, Expo, TypeScript, and Firebase.
- Developed TypeScript Cloud Functions and Firestore workflows for quest completion and user progression, using Firestore transactions to ensure data integrity and prevent duplicate processing.
- Deployed an LLM validation pipeline with schema/difficulty classification and feedback logging, evaluated on 100+ human-labeled quests at 82% agreement and 94% violation catch rate.

The Illinois Space Society - *Software Engineer* Sept 2024 – May 2025
- Developed data-processing tools to analyze rocket telemetry in real time, enabling faster performance diagnostics.
- Integrated Kalman filters for position and velocity estimation, improving accuracy of flight path reconstruction.
- Collaborated with avionics and propulsion teams to ensure reliable data pipelines and mission readiness.

Projects

Real-Time Emotion Recognition System | *Python, TensorFlow, Keras, OpenCV, React, Firebase, Twilio*
- Trained a TensorFlow/Keras emotion recognition model on 35K+ labeled images to classify facial expressions.
- Constructed an OpenCV pipeline to process live video frames and run real-time emotion inference with the model.
- Built a React/Firebase monitoring workflow with Twilio to track emotional patterns and alert concerning activity.

Autonomous Drone Controller | *Python, PyTorch, TensorFlow*
- Developed custom controller software to autonomously stabilize and navigate a quadrotor drone.
- Designed PID altitude and LQR attitude controllers, using Lyapunov stability to verify performance.
- Trained a neural network controller in simulation to optimize flight stability and control gains.
- Produced graphs and visualizations from simulation data to analyze control response and performance.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C++, C, SQL, R, Swift, HTML/CSS, Assembly (MIPS)
Frameworks/Libraries: React, Next.js, React Native, Node.js, PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy
Tools/Technologies: Git/GitHub, Docker, Firebase, Google Cloud, MongoDB, REST APIs, Jira, Vite, Xcode